CMP5366 Coursework Task Sheet 2
Data Analytics Pipeline: Application and Analysis

Your third task will be to **access** and **critically evaluate the quality** of your implemented Machine Learning pipeline. This process must be clearly documented in your **report**.

The purpose of this task is to contribute towards meeting the following module learning outcomes:

- "Appraise the quality of the implemented Machine Learning pipeline by applying appropriate techniques and examining environmental sustainability and social responsibility."
- "Analyse and communicate the ethical and legal issues that concern security, privacy, and general data processing in Data Science environments."

Underlined are some of the key parts of the learning outcome, but what do they mean?

- Appraise the quality of the implemented Machine Learning pipeline
  - In this task, you should critically evaluate how well your pipeline performs. Consider its efficiency, scalability, accuracy, maintainability, and overall effectiveness. Think about what went well, what didn't, and what you might change to improve it.
- Appropriate techniques
  - You will need to use suitable tools and methods to support your evaluation, such as performance metrics, visualisation, data analysis, or logging techniques. You should also revisit outputs from earlier stages (e.g., exploratory data analysis, model training) to inform your appraisal.
- Environmental sustainability and social responsibility
  - Reflect on the environmental impact of your pipeline. Could any part of your process (e.g., training, storage, or data handling) be made more energy efficient or less resource intensive.
- Analysis and communicate
  - Demonstrate the ability to judge a given concept, technique or event backed by appropriate usage of evidence
- Ethical
  - What moral implications are there behind how data may be used?
- Legal
  - What legislation exists to attempt to limit the inappropriate handling of data?
- Security
  - How can data and the systems that process data be protected from unauthorised access and usage?
- Privacy
  - How does the manner that data is handled effectively allow people to control how personal information is handled?

What you need to do

Exercise 2: Research based discussion

Step 1 - Pipeline Quality

Critically evaluate the quality of your implemented Machine Learning pipeline by answering the following:

- What tools or techniques did you use to assess your pipeline?
  - e.g., Did you check performance metrics or errors in the model output?
- Could your pipeline be more environmentally friendly?
  - e.g., Could you simplify steps or reduce repetition?
- Is your pipeline socially responsible?
  - e.g., Is your data fair and unbiased?

Step 2 – Drift

Using independent research and academic literature **(provide references)**, explore and discuss:

- How could **drift** happen in *your* project?
- How can **model drift** be detected and managed in a **deployed, production Machine Learning** system?

Step 3 – Evaluation and Communication of Wider issues

Using independent research and academic literature (provide references), explore and discuss:

- According to **data protection regulations**, what **ethical** and **legal** concerns are most relevant to how you are currently handling and processing data within your pipeline?
- Within your prototype system, what steps could you take to enhance the **security** and **privacy** of your data analytics pipeline?

Marking

Marks will be awarded within this task based on:

- Providing a **sufficiently detailed**, **research-supported,** discussion of the above topics.
- A **reflective discussion** of the process undertaken including:
  - How would you improve the structure of the pipeline in the future, specifically regarding the concepts discussed within this task sheet?
  - What have you learned from **undertaking the process**? (technical skills, knowledge and more general skills)